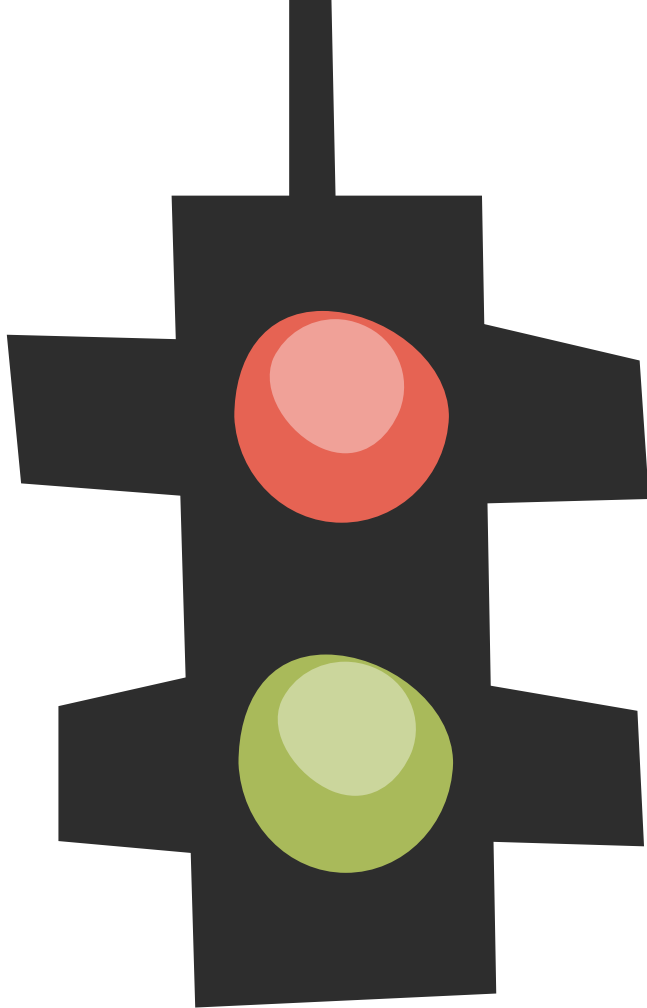




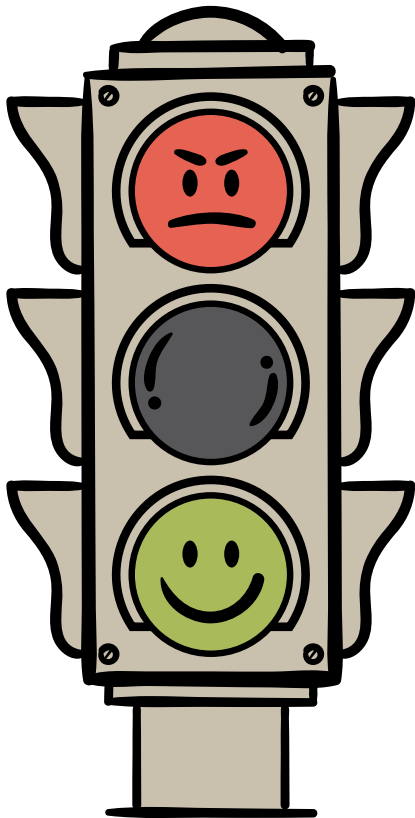
Red Light, Green Light Using Embedded Computer Vision

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CS 4391.001

Purpose of the Project

Goal: Build an automated referee for Red Light, Green Light using pose estimation.



The Problem

- Human motion can be hard to judge automatically.
- Small body sway and camera noise can cause false movement detections.
- The system must track two players in real time.



Why It Matters

- Demonstrates real-time human pose estimation.
- Applies computer vision to interactive games.
- Can extend to fitness, physical therapy, safety monitoring, sports, or classroom activities.

How the Game Works



System Overview

01 Camera Input

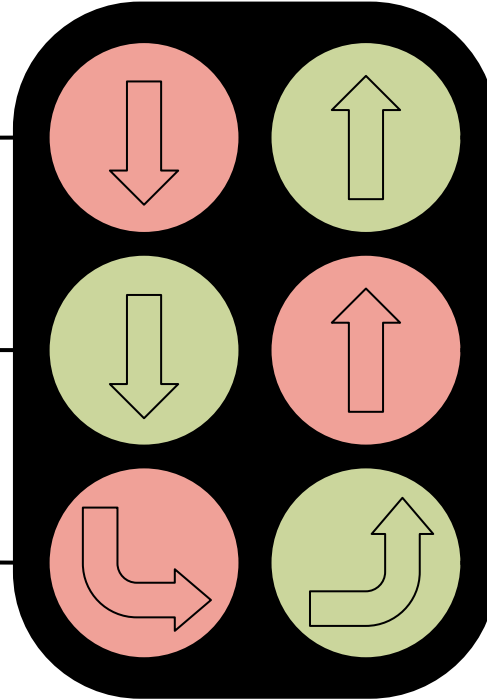
Webcam captures live frames.

02 YOLOv8-Pose

YOLOv8-Pose detects human keypoints.

03 Keypoints

Keypoints are smoothed to reduce jitter.



04 Player Tracking

System assigns players to Player 1 and Player 2.

05 Motion Logic

Motion is measured during Red Light. Stillness is measured during Green Light.

06 Game Output

Game logic decides: strike, eliminated, or winner.

Computer Vision Approach



Model

- Used pretrained YOLOv8-Pose.
- Detects human body keypoints in each frame.
- Keypoints include head, shoulders, elbows, wrists, hips, knees, and ankles.

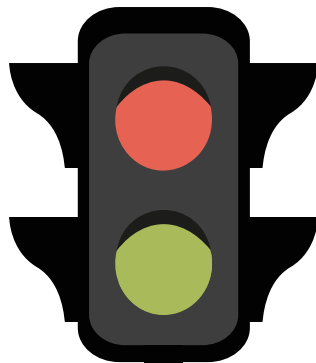
Motion Detection

- We compare the player's current pose to their red-light baseline.
- Keypoints are smoothed over time to reduce jitter.
- Movement is normalized by body size so distance from camera is handled better.

Multi-Player Detection

The Problem

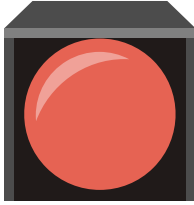
- YOLO can detect multiple people.
- If players cross lanes, simple lane tracking can create Player 3 or Player 4.
- Overlapping players can cause ID switching.



The Solution

- Lock the game to Player 1 and Player 2.
- Match detections using body center, pose similarity, and body scale.
- Do not create new players after setup.
- Lane crossing does not create a new ID.

Implementation



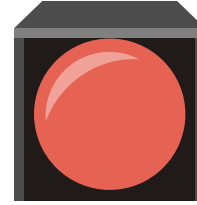
Tools

Python
OpenCV
Ultralytics
YOLOv8-Pose
Webcam / Jetson
Orin Nano target
platform



Main Components

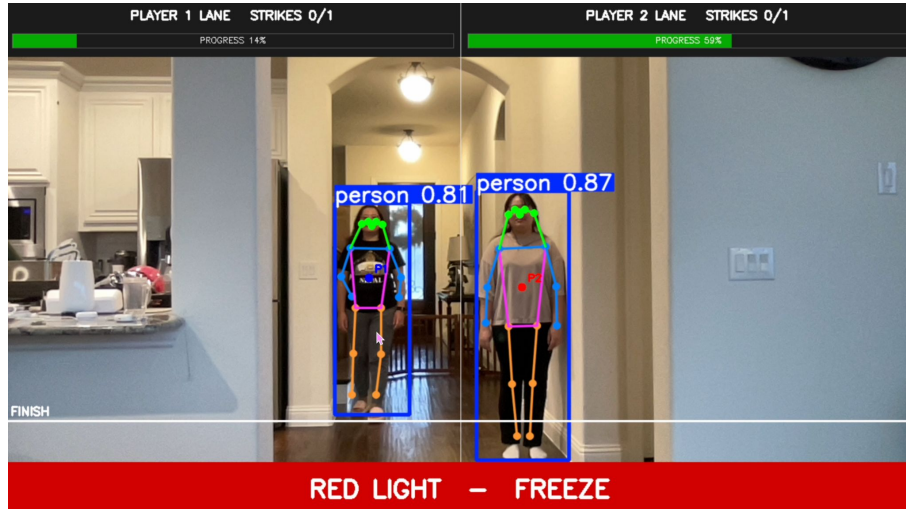
Camera capture and video
recording
Pose keypoint extraction
Player tracking
Red/green timer logic
Keypoint smoothing
Normalized movement scoring
Strike and elimination logic



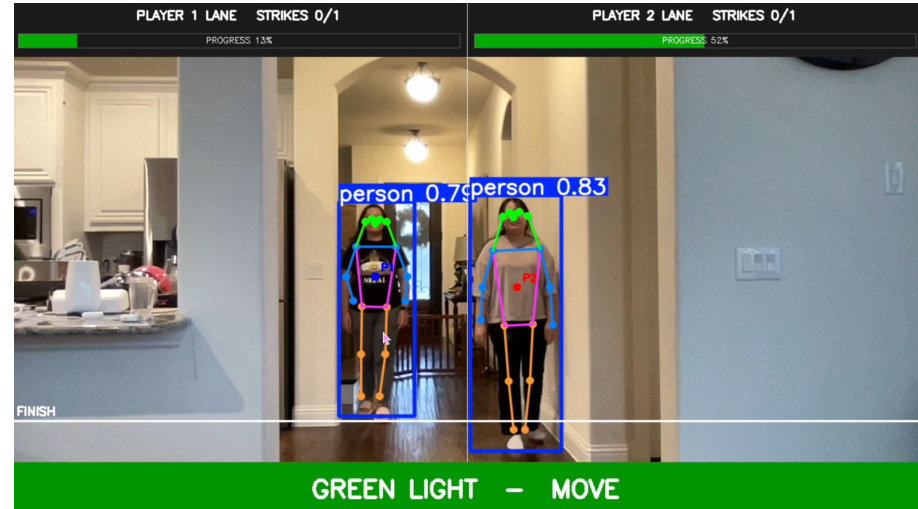
Latest Improvements

Less sensitive to natural
sway
More stable two-player
tracking
No Player 3 / Player 4
issue
Separate analysis script
for dataset metrics

Demo

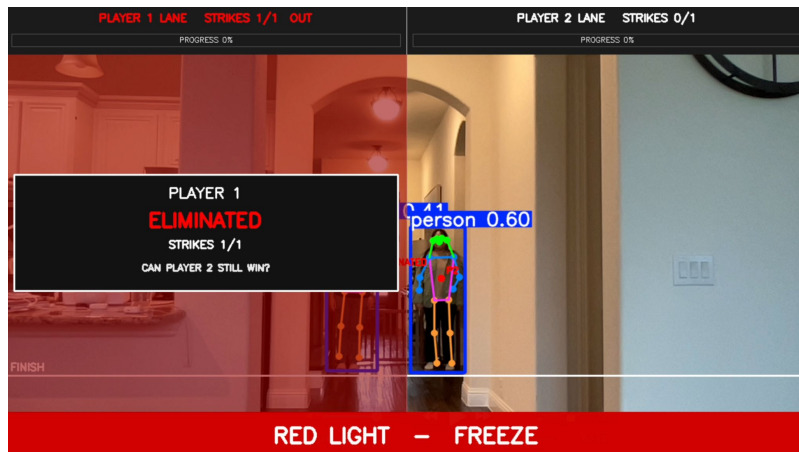


Red Light - Players cannot move

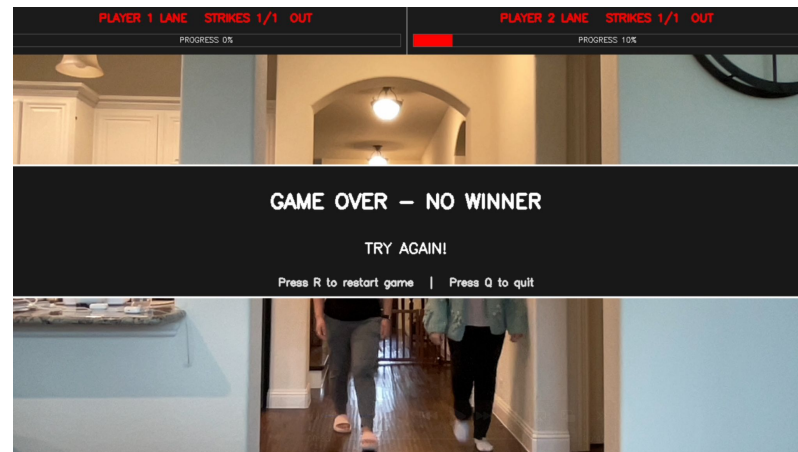


Green Light - Players can move to complete the progress bar

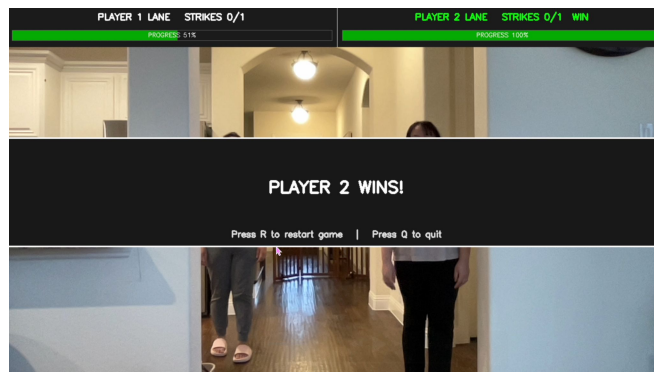
Demo



Move during Red Light : **Eliminated**



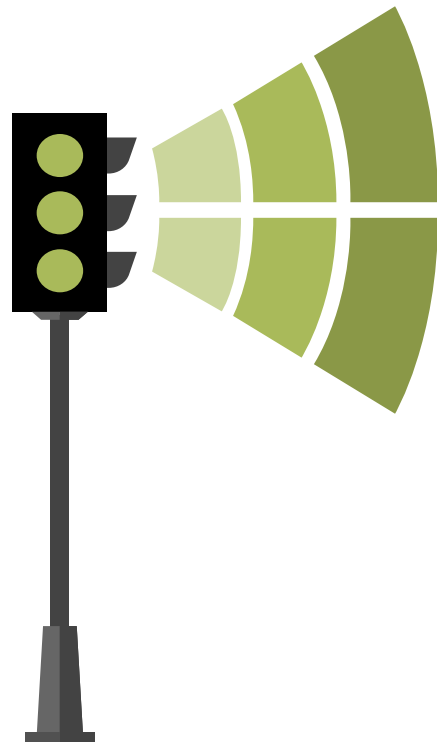
Both player move: **Game Over**



Progress bar complete: **Player wins!**

Results

Metric	Midterm Prototype	Final System
Video-level accuracy	80.0%	92.9%
Accuracy improvement	—	+12.9%
Still false-positive rate	11.9%	6.4%
Large-motion frame detection rate	76%	97.1%
Dataset size	18 clips / 1,754 frames	34 clips / 2,218 frames
Laptop / Jetson runtime	13 FPS / 8 FPS	30 FPS / 15 FPS



Challenges

Pose Jitter

YOLO keypoints move slightly even when the person is standing still.

Two-Player Tracking

Players can overlap or move into each other's lanes.

Small Natural Movement

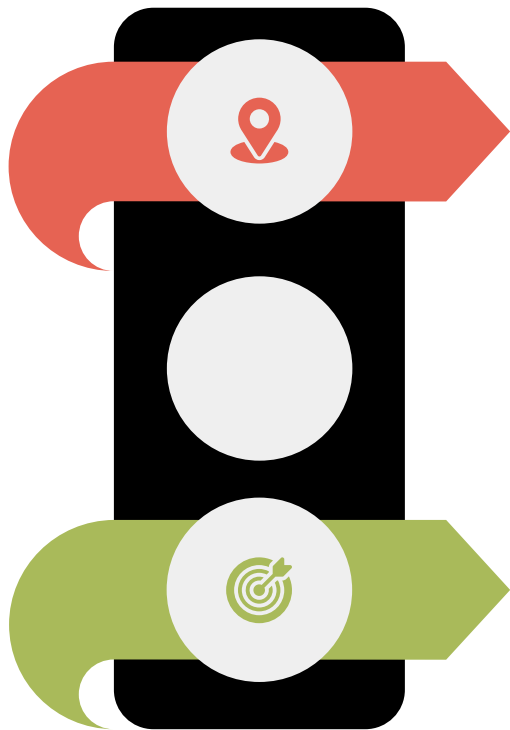
Real players sway, shift weight, or move their arms slightly.

Embedded Performance

TensorRT optimization improved Jetson performance but required extra setup and validation.



Improvements & Future Work



Future Improvements

- Add more evaluation videos with more players and lighting, including outdoors.
- Add better finish-line/depth estimation.
- Add automatic calibration for different camera positions.
- Test with more gameplay scenarios/activities.

Takeaway

- A pretrained pose model plus stable post-processing can create a real-time automated Red Light, Green Light referee.